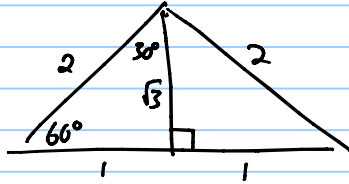


$$\sin(\theta) = \frac{y}{r}$$

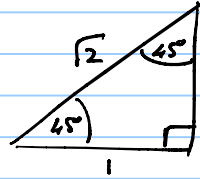
$$\cos(\theta) = \frac{x}{r}$$

$$\tan(\theta) = \frac{y}{x}$$

Special angles (know these, see study guide!)



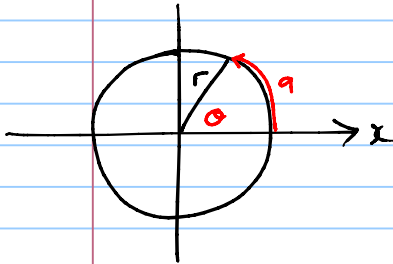
$$\Rightarrow \sin 60^\circ = \frac{\sqrt{3}}{2}$$



$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

In Calculus, we don't work with degrees, but with RADIANS (another way to measure an angle).

Consider the circle with radius r .



The arc length (a) is proportional to the size of the angle (θ).

$$\therefore a \propto \theta$$

$$\text{We say: } a = r\theta$$

$$\Rightarrow \theta = \frac{a}{r} \quad \left(\frac{\text{length}}{\text{length}} \right)$$

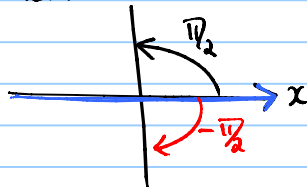
r = radius of circle.

When $\theta = 360^\circ$, $a = 2\pi r$ (circumference of circle),

$$\text{Hence: } 360^\circ = \frac{2\pi r}{r} = 2\pi \text{ radians}$$

$$\text{and } 180^\circ = \pi \text{ radians.}$$

- Know the table on p. A24.
- We measure angles from the standard position: see p. A25.

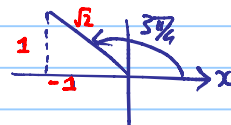
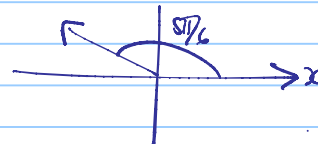


- Trig. functions: see p. A26, know the table on p. A27.

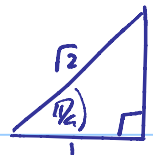
Examples:

$$\begin{aligned} 1) \quad \sin\left(\frac{5\pi}{6}\right) &= \sin\left(\pi - \frac{\pi}{6}\right) \\ &= \sin\left(\frac{\pi}{6}\right) = +\frac{1}{2} \end{aligned}$$

$$\begin{aligned} \cos\left(\frac{3\pi}{4}\right) &= \cos\left(\pi - \frac{\pi}{4}\right) = -\cos\frac{\pi}{4} \\ &= -\frac{1}{\sqrt{2}} \end{aligned}$$

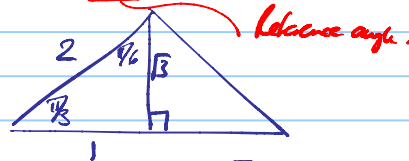
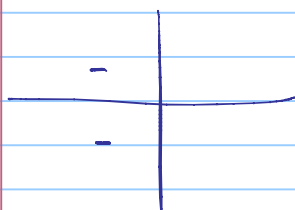


$$\begin{aligned}\tan\left(\frac{9\pi}{4}\right) &= \tan\left(\frac{8\pi + \pi}{4}\right) = \tan\left(2\pi + \frac{\pi}{4}\right) \\ &= \tan \frac{\pi}{4} \\ &= 1.\end{aligned}$$



2) Determine x if $\cos x = -\frac{\sqrt{3}}{2}$, given $x \in [0, 2\pi]$.

$\cos x$ is negative if x is in 2nd or 3rd quadrant, and: $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$.

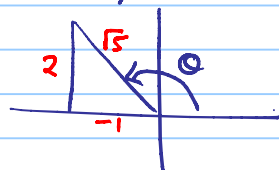


$$\begin{aligned}\Rightarrow x &= \pi - \frac{\pi}{6} \quad \text{or} \quad x = \pi + \frac{\pi}{6} \\ &= \frac{5\pi}{6} \quad \quad \quad = \frac{7\pi}{6} \\ \Rightarrow x &\in \left\{ \frac{5\pi}{6}, \frac{7\pi}{6} \right\}.\end{aligned}$$

3) If $\tan \theta = -2$, $\frac{\pi}{2} < \theta < \pi$, find $\cos \theta$.

$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{2}{-1}$ and θ in second quadrant.

$$\Rightarrow \cos \theta = \frac{-1}{\sqrt{5}}$$



• Trig reciprocals:

$$\begin{aligned}\operatorname{cosec}(x) &= \frac{1}{\sin(x)} \\ \sec(x) &= \frac{1}{\cos(x)} \\ \cot(x) &= \frac{1}{\tan(x)}\end{aligned}$$

Ex: Find all the trigonometric ratios of $\frac{7\pi}{4}$.

$$\sin\left(\frac{7\pi}{4}\right) = \sin\left(2\pi - \frac{\pi}{4}\right) = -\sin \frac{\pi}{4} = -\frac{1}{\sqrt{2}}.$$

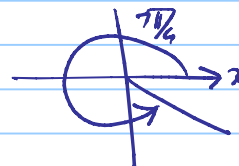
$$\operatorname{cosec}\left(\frac{7\pi}{4}\right) = -\sqrt{2}.$$

$$\cos\left(\frac{7\pi}{4}\right) = \cos\left(2\pi - \frac{\pi}{4}\right) = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sec\left(\frac{7\pi}{4}\right) = \sqrt{2}. \quad (\sec \alpha = \frac{1}{\cos \alpha})$$

$$\tan\left(\frac{7\pi}{4}\right) = \tan\left(2\pi - \frac{\pi}{4}\right) = -\tan \frac{\pi}{4} = -1$$

$$\Rightarrow \cot\left(\frac{7\pi}{4}\right) = -1.$$



• See examples 3, 4 on p. A27.

• Trig identities: p. A28, A29 - see study guide!

Example:

i) Prove that $2\operatorname{cosec}(2t) = \sec(t)\operatorname{cosec}(t)$

We start with simplifying the left hand side:

$$\text{LHS} = 2\operatorname{cosec}(2t) = \frac{2}{\sin(2t)} = \frac{2}{2\sin(t)\cos(t)}$$

$$= \frac{1}{\sin(t)\cos(t)}$$

$$= \operatorname{cosec}(t) \sec(t) = \sec(t) \operatorname{cosec}(t).$$

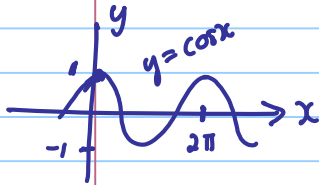
$$= \underline{\text{RHS.}}$$

2) Example 7, p. A30

3) Solve for x if $\operatorname{cosec}(x) = \cot(x)$.

$$\operatorname{cosec}(x) = \cot(x) \Rightarrow \frac{1}{\sin(x)} - \frac{\cos(x)}{\sin(x)} = 0$$

$$\Rightarrow \frac{1 - \cos(x)}{\sin(x)} = 0$$



$$\Rightarrow 1 - \cos(x) = 0 \quad \text{and } \underline{\sin(x) \neq 0}$$

$$\Rightarrow \cos(x) = 1$$

$$\Rightarrow x = 0, 2\pi, 4\pi, \dots, \quad \text{but } \sin(x) = 0 \quad \text{for } x = 2k\pi$$

$$\Rightarrow x \in \emptyset.$$